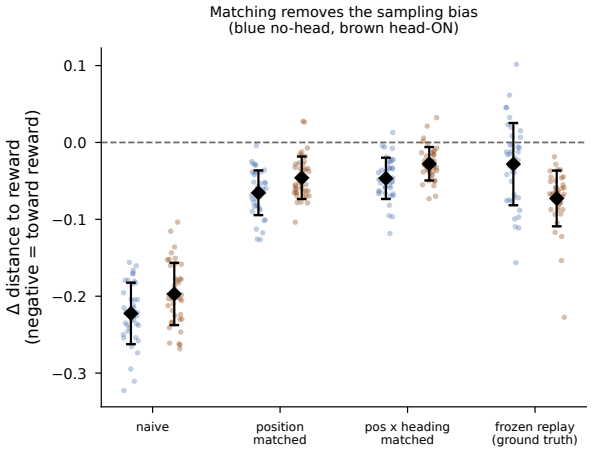
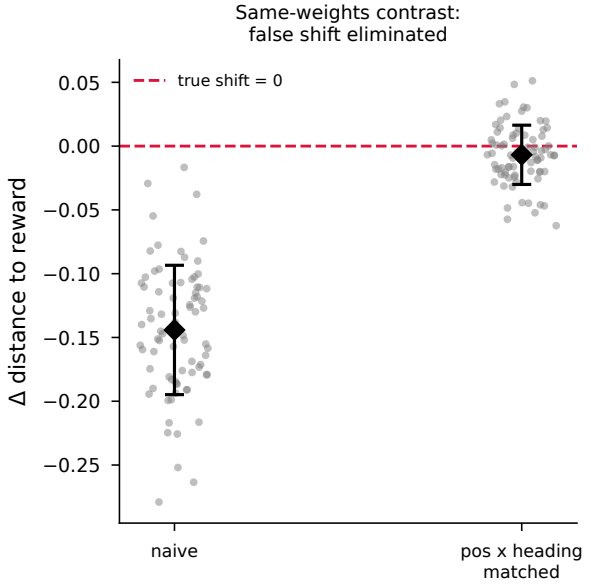
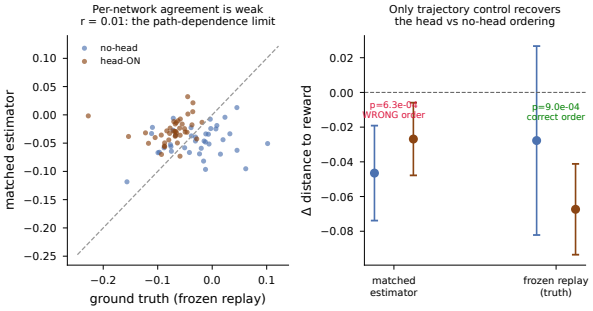
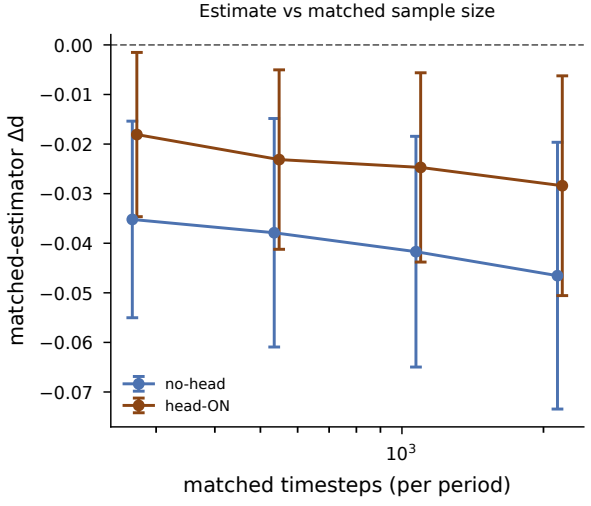

Matching removes the sampling bias
(blue no-head, brown head-ON)
Δ distance to reward
(negative = toward reward)
0.1
0.0
−0.1
−0.2
−0.3
naive
position
matched
pos x heading
matched
frozen replay
(ground truth)


Same-weights contrast:
false shift eliminated
true shift = 0
Δ distance to reward
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
naive
pos x heading
matched


Per-network agreement is weak
r = 0.01: the path-dependence limit
no-head
head-ON
matched estimator
ground truth (frozen replay)
Only trajectory control recovers
the head vs no-head ordering
Δ distance to reward
p=6.3e-01
WRONG order
p=9.0e-04
correct order
matched
estimator
frozen replay
(truth)


Estimate vs matched sample size
matched-estimator Δd
0.00
−0.01
−0.02
−0.03
−0.04
−0.05
−0.06
−0.07
no-head
head-ON
10^3
matched timesteps (per period)